

**Non-Markovian gravitational decoherence from indivisible
stochastic processes: a Gaussian-onset alternative to
Diósi–Penrose and a fit-free interferometric test**

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Abstract

Gravitationally induced decoherence, in the Diósi–Penrose (DP) form, predicts that a spatial superposition of a massive body loses coherence at a rate set by the gravitational self-energy E_G of the mass-density difference between the branches, with characteristic time $\tau_G = \hbar/E_G$. The DP law is exponential in time, a form that follows from an implicit Markov (memoryless, divisible) assumption about the underlying stochastic dynamics. We examine gravitational decoherence within the framework of *indivisible stochastic processes* (ISP) of Barandes, whose defining feature is the failure of the divisibility (Chapman–Kolmogorov) property. We show that this single structural feature generically replaces the DP exponential with a non-exponential coherence decay whose short-time behaviour is Gaussian, $C(T) = C(0) \exp[-\frac{1}{2}(T/\tau_G)^2]$, at the *same* energy scale E_G , crossing over to an exponential tail beyond a record-memory time τ_c . The Markovian DP law is recovered as the $\tau_c \rightarrow 0$ limit. The distinction is experimentally sharp and requires no fitted constant: the *onset shape* of the residual gravitational log-coherence is linear in T for Markovian collapse (finite initial slope) and quadratic (zero initial slope, a Zeno-like plateau) for the indivisible case. We give threshold masses and separations for levitated-nanosphere, matter-wave, and mechanical-cat-state platforms, analyse what fixes τ_c , and address the spontaneous-radiation bound that already excludes the parameter-free DP model. We are explicit about scope: the quadratic onset is the generic signature of non-Markovian decoherence and is shared with other coloured-noise collapse models, so the test falsifies *strict* Markovian DP/CSL and confirms a non-Markovian fundamental channel consistent with ISP, but does not by itself single out ISP.

I. INTRODUCTION

The persistence of quantum coherence for increasingly massive systems is one of the sharpest open questions at the quantum–classical and quantum–gravity interfaces. Objective-collapse models posit a fundamental, observer-independent mechanism that suppresses macroscopic superpositions [4–7]. Among them, the proposals of Diósi [2, 3] and Penrose [1] tie this suppression specifically to gravity: a superposition of two distinct mass distributions is a superposition of two distinct spacetime geometries, and the gravitational self-energy of their difference sets a decoherence time $\tau_G = \hbar/E_G$. This “Diósi–Penrose” (DP) decoherence is now an active experimental target, both directly through matter-wave and optomechanical

cal interferometry [16, 19, 24] and indirectly through spontaneous-radiation searches [8, 9], the latter having already excluded the simplest, parameter-free DP model [8].

The DP coherence law is exponential, $C(T) = C(0) \exp(-T/\tau_G)$. This functional form is not an independent prediction of gravity; it is a consequence of treating the underlying collapse noise as *white* (delta-correlated in time), equivalently of assuming the reduced dynamics forms a Markovian semigroup. Real physical noise is never white [10], and the open-systems literature shows that non-Markovian (coloured) environments generically produce non-exponential coherence decay, with a universal quadratic (Zeno-like) onset at short times [15, 21, 22].

In this paper we ask what gravitational decoherence looks like when the underlying dynamics is taken to be *indivisible* in the sense of Barandes' stochastic-quantum correspondence [11, 12], building on the older program of recovering quantum mechanics from stochastic processes [13]. An indivisible stochastic process is one whose transition law does *not* factorize through intermediate times,

$$\Gamma(t_2, t_0) \neq \Gamma(t_2, t_1) \Gamma(t_1, t_0), \quad (1)$$

i.e. it violates exactly the divisibility property that underlies the Markovian semigroup. Our central result is that this single feature replaces the DP exponential with a non-exponential law whose short-time form is Gaussian, at the same energy scale, and that the difference is experimentally accessible through a fit-free onset-shape measurement. We recover DP as the Markovian limit, identify the new time scale τ_c that controls the crossover, and confront the existing spontaneous-radiation bound.

Our claims are deliberately bounded. The mechanism is a *reconstruction* of gravitational decoherence within a particular stochastic ontology; the new, falsifiable content is the non-exponential onset, not the energy scale (which is DP's). The quadratic onset is the generic signature of non-Markovianity and is not unique to ISP. We make these limitations explicit in Sec. IX.

II. INDIVISIBLE STOCHASTIC DYNAMICS AND THE RECORD-STABILIZATION HYPOTHESIS

Barandes’ stochastic-quantum correspondence [11, 12] establishes that a broad class of quantum systems can be represented as stochastic processes over a configuration space in which the configuration is a definite quantity (“beable”) at all times, with the unitary, Hilbert-space description emerging as an exact reformulation. The processes that correspond to generic quantum evolution are *indivisible*: their transition matrices do not satisfy the Chapman–Kolmogorov composition law except at isolated *division events*, times at which the process momentarily factorizes and behaves Markovianly. Division events are the natural locus of effective measurement and record formation.

We adopt the following hypothesis, consistent with this picture and with the gravitational-decoherence proposals [1–3, 20]: when two coherent branches carry distinct effective mass-density records, the gravitational distinguishability of the corresponding geometries acts as a channel that *stabilizes* the branches into distinct records. The stabilization is not a primitive collapse postulate; it is the statement that the gravitational mismatch supplies a positive, accumulating “cost” that suppresses the survival of the unstabilized (coherent) alternative.

We distinguish three physically separate effects, only the third of which is the subject of this paper: (i) *environmental decoherence*—ordinary scattering, gas collisions, thermal emission, reducible in principle by better isolation [14, 16]; (ii) *unitary gravitational phase*—gravity shifts the relative phase of the branches without destroying coherence; (iii) *gravitational record stabilization*—the geometric mismatch itself acts as a branch-stabilizing (decohering) channel. Effect (iii) is the putative new physics.

III. GRAVITATIONAL MISMATCH ENERGY AND THE MARKOVIAN LIMIT

Let the two branches carry mass densities $\rho_r(\mathbf{x})$ and $\rho_{r'}(\mathbf{x})$, with difference $\delta\rho = \rho_r - \rho_{r'}$. Following Diósi and Penrose [1–3], the gravitational mismatch energy is the Newtonian self-energy of the difference,

$$E_G = \frac{G}{2} \int d^3x d^3y \frac{\delta\rho(\mathbf{x}) \delta\rho(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}. \quad (2)$$

E_G is non-negative once a finite object size is retained; the point-mass limit is not admissible because it diverges, exactly the regularization issue that the experimental bounds constrain

[8]. Define the dimensionless gravitational action accumulated over a hold time T ,

$$a(T) = \frac{E_G T}{\hbar}, \quad \tau_G = \frac{\hbar}{E_G}. \quad (3)$$

The standard DP result is the exponential coherence law $C(T) = C(0) \exp(-T/\tau_G)$.

It is instructive to see precisely which assumption produces the exponential. Write the survival factor $S(E_G, T) = |C(T)|/|C(0)|$. If one assumes (i) time multiplicativity $S(E_G, T+U) = S(E_G, T) S(E_G, U)$, (ii) additivity of independent cost channels $S(E_1 + E_2, T) = S(E_1, T) S(E_2, T)$, and (iii) dependence only on the dimensionless action a , then the Cauchy functional equation forces $S = \exp(-\kappa a) = \exp(-\kappa E_G T/\hbar)$, with the coefficient κ fixed by normalization. Assumption (i) is the semigroup (Chapman–Kolmogorov) property: it states that processing over the whole interval factorizes into independent processing over its parts. *This is the divisibility / Markov assumption.* The DP exponential is therefore the Markovian special case of a more general survival functional. We now drop assumption (i), as the indivisible structure requires.

IV. INDIVISIBLE DYNAMICS AND NON-EXPONENTIAL DECOHERENCE

A. The general survival functional

For a process that is not divisible, the survival factor is not multiplicative in time and need not be exponential. The model-independent object is the decoherence functional [15, 21]

$$\ln \frac{|C(T)|}{|C(0)|} = -\Gamma(T), \quad \Gamma(T) = \frac{1}{2\hbar^2} \int_0^T \int_0^T K(t_1 - t_2) dt_1 dt_2, \quad (4)$$

where $K(s) = \text{Cov}[\delta E(t), \delta E(t+s)]$ is the autocorrelation of the gravitational record-mismatch energy, with fluctuation amplitude σ_E and correlation (record-memory) time τ_c . Two limits bracket the behaviour.

Markovian limit ($\tau_c \rightarrow 0$, $K \propto \delta$): the double integral is linear in T , $\Gamma(T) \propto T$, recovering the DP exponential and identifying $1/\tau_G = E_G/\hbar$.

Maximally indivisible limit ($\tau_c \rightarrow \infty$, no internal decorrelation—the natural meaning of a single indivisible transition over the hold): $K \approx \sigma_E^2$ is constant, so $\Gamma(T) = \frac{1}{2}(\sigma_E T/\hbar)^2$, a quadratic exponent and hence a Gaussian coherence decay.

B. The central result

Taking the natural reading in which the relative which-record amplitude is held coherently across a single indivisible transition, so that σ_E is of order E_G , the shape is fixed and the scale is the *same* τ_G as DP:

$$\boxed{C(T) = C(0) \exp\left[-\frac{1}{2} (T/\tau_G)^2\right], \quad \tau_G = \hbar/E_G.} \quad (5)$$

For a generic finite τ_c , with an exponential memory kernel $K(s) = \sigma_E^2 e^{-|s|/\tau_c}$ one obtains the closed form

$$\Gamma(T) = \frac{\sigma_E^2 \tau_c^2}{\hbar^2} \left(\frac{T}{\tau_c} - 1 + e^{-T/\tau_c} \right), \quad (6)$$

which is Gaussian for $T \ll \tau_c$ and crosses over to a linear (exponential) tail for $T \gg \tau_c$. The DP exponential is the $\tau_c \rightarrow 0$ edge; the pure Gaussian is the $\tau_c \rightarrow \infty$ edge. This is the quantitative content of indivisibility for gravitational decoherence: same energy scale, non-exponential shape, with a single new time scale τ_c controlling the crossover.

The short-time quadratic onset is not an artefact of the exponential kernel; it is the universal consequence of a finite-correlation-time (smooth) noise, identical in origin to the quantum-Zeno short-time behaviour of any open system [22], and is required by the analyticity of $\Gamma(T)$ together with $\Gamma'(0) = 0$ whenever the noise has no white (delta) component.

V. A FIT-FREE INTERFEROMETRIC DISCRIMINATOR

The practical signature is the *shape* of the decay near $T = 0$, which carries no fitted constant. Prepare a spatial superposition with fixed gravitational mismatch E_G and a fixed environmental budget Γ_{env} , and measure the interferometric visibility $V(T)$. Define the residual gravitational log-coherence by subtracting the separately measured environmental contribution (e.g. by repeating at $E_G \rightarrow 0$, i.e. negligible separation),

$$L(T) = \ln \frac{V(T)}{V_{\text{env}}(T)}. \quad (7)$$

The three hypotheses give qualitatively distinct onsets,

$$L(T) = \begin{cases} 0, & \text{ordinary QM (no fundamental channel),} \\ -T/\tau_G, & \text{Markovian DP/CSL: finite slope,} \\ -\frac{1}{2}(T/\tau_G)^2, & \text{indivisible ISP: zero slope.} \end{cases} \quad (8)$$

Gravitational coherence decay: Markovian (DP) exponential vs indivisible (ISP) Gaussian

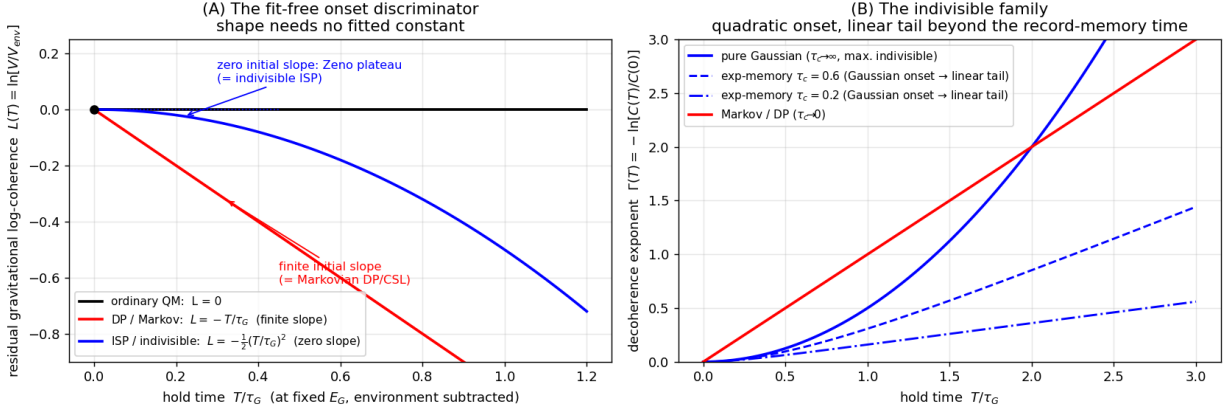


FIG. 1. (a) The fit-free onset discriminator: residual gravitational log-coherence $L(T)$ at fixed E_G . Ordinary QM is flat; Markovian DP/CSL is linear with a finite initial slope; the indivisible (ISP) channel is quadratic, tangent to the axis at the origin (a Zeno-like plateau). (b) The indivisible family of Eq. (6): quadratic onset crossing over to a linear tail beyond the record-memory time τ_c ; the Markovian DP exponent is the $\tau_c \rightarrow 0$ edge.

The discriminator is therefore: a straight line through the origin implies Markovian DP/CSL; a parabola tangent to the axis at the origin implies the indivisible channel (Fig. 1). Because the *shape* (linear vs quadratic onset) is independent of the overall coefficient, the test does not require knowing τ_G in advance, nor the absolute collapse strength—only the ability to vary E_G while holding the environment fixed, and to resolve the curvature of $L(T)$ at small T .

A second axis: distinguishing ISP from the whole collapse-model family. The onset shape separates ISP from any *Markovian* collapse model, but several related theories must be told apart at once, and they are distinguished by *two* axes — onset shape and rate scaling. (Both axes are plain, non-relativistic ISP: indivisibility supplies the onset and the Newtonian self-energy E_G supplies the scaling; the relativistic extension adds nothing here.) ISP occupies a cell no other named theory does:

A two-axis measurement settles it. *Axis 1 (onset shape)* separates ISP from the Markovian models (exponential vs Gaussian). *Axis 2 (rate scaling)*: vary M , d , and geometry and test whether the rate tracks the gravitational self-energy E_G (the Diósi–Penrose form, $\propto G$) or the CSL parameters (λ, r_C , a fixed $\sim 10^{-7}$ m length scale and a different geometry

theory	decay?	onset	rate scaling
unitary QM/QFT	no	—	—
Schrödinger–Newton [23]	no (phase shift)	—	gravitational, but a shift
CSL/GRW [4, 5]	yes	exponential	own (λ, r_C) , not E_G
Diósi–Penrose [1, 2]	yes	exponential	gravitational E_G
ISP (this work)	yes	Gaussian	gravitational E_G

dependence); Schrödinger–Newton appears instead as a deterministic frequency *shift* with no decay, already bounded by optomechanics [23]. The (Gaussian onset) $\times$ (gravitational E_G -scaling) cell points at ISP and excludes Markovian DP (exponential), CSL/GRW (wrong scaling), Schrödinger–Newton (shift, not decay), and unitary QM (no decay) at once.

Honest limits. The one construction sharing the ISP cell is a non-Markovian *gravitational* collapse — ISP’s own content under another name, not a distinct theory; a colored CSL [10] can mimic the Gaussian onset but not the E_G -scaling, which the second axis separates. Axis 2 is a campaign (map the E_G -scaling across M, d , geometry); the test presupposes the channel exists (else every row is “no decay” and ISP is empirically QM); and the regime is at the frontier of mesoscopic-superposition sensitivity. Within these limits the two-axis test upgrades the discriminator from “ISP vs. quantum mechanics” to “ISP vs. the entire collapse/decoherence family.”

VI. NUMERICAL THRESHOLDS AND EXPERIMENTAL PLATFORMS

We quote thresholds for a rigid homogeneous sphere of mass M , radius R , density ρ , coherently separated by d . For $d \ll R$ the mismatch energy is harmonic,

$$E_G \simeq \frac{1}{2} M \omega_G^2 d^2, \quad \omega_G^2 = \frac{4\pi G \rho}{3}, \quad (9)$$

while for object-scale separations it saturates at the self-energy scale $E_G^{\text{sat}} \simeq \frac{6}{5} G M^2 / R$. Using $\rho = 2000 \text{ kg m}^{-3}$, the saturated threshold mass that decoheres in time T (i.e. $\tau_G = T$) is

$$M_{\text{thr}}^{\text{sat}}(T) = \left[\frac{\hbar}{\frac{6}{5} G (4\pi \rho / 3)^{1/3} T} \right]^{3/5}. \quad (10)$$

Representative values (saturated regime, $d \gtrsim R$) are given in Table I. These mass and separation windows coincide with the targets of current and proposed mesoscopic-superposition

TABLE I. Saturated gravitational-decoherence thresholds for a homogeneous sphere of density $\rho = 2000 \text{ kg m}^{-3}$: mass that gives $\tau_G = T$, and the corresponding object radius.

T	M_{thr} (kg)	M_{thr} (amu)	R_{thr}
1 μs	3.1×10^{-12}	1.9×10^{15}	7.2 μm
1 ms	4.9×10^{-14}	2.9×10^{13}	1.8 μm
1 s	7.7×10^{-16}	4.6×10^{11}	452 nm
100 s	4.9×10^{-17}	2.9×10^{10}	180 nm
1 h	5.7×10^{-18}	3.4×10^9	88 nm

experiments [16, 17, 19, 24].

Three platform classes are natural hosts for the onset-shape test: (i) *levitated nanosphere interferometry*—neutral dielectric nanospheres in ultra-high vacuum, prepared in spatial superposition and recombined [16]; (ii) *mesoscopic matter-wave interferometry*—large molecule and cluster interferometers, where macroscopicity has been advanced systematically [16, 24]; (iii) *mechanical cat states / optomechanics*—superpositions of a massive mechanical element [19], including the gravity-mediated-entanglement geometries [17, 18] in which E_G can be tuned by separation. In every case the strategy is identical: hold the object, temperature, vacuum, and pulse sequence fixed, vary E_G (e.g. through the separation d), and examine whether the gravitational contribution to $\ln V$ enters linearly or quadratically in the hold time.

VII. THE RECORD-MEMORY TIME

The new scale τ_c is the autocorrelation time of the gravitational mismatch—in the ISP language, the mean spacing between division events of the gravitational record channel. A self-consistency argument fixes its order: in the present setting the division event *is* the record stabilization, and the stabilization time is τ_G , so the memory time and the stabilization time coincide, $\tau_c \sim \tau_G = \hbar/E_G$.

This identification is dimensional rather than unique. The candidate scales are $\hbar/E_G = \tau_G$ (channel-intrinsic; division = stabilization), placing the Gaussian onset within the observable window $T \sim \tau_G$; $(3/4\pi G\rho)^{1/2} = 1/\omega_G$ (the gravitational dynamical time, of order 10^3 s for

solid-density material), also observable; and R/c (the light-crossing time, of order 10^{-15} s), which would Markovianize the channel and recover the DP exponential. The honest verdict is that indivisibility *forces* the decay to be non-exponential but pins τ_c only up to which physical scale governs the channel; the channel-intrinsic and gravitational-dynamical choices both place the crossover within reach of the experiments of Sec. VI, while only a fast microphysical floor would restore DP.

VIII. SPONTANEOUS RADIATION AND EXISTING BOUNDS

The parameter-free DP model is experimentally excluded. Donadi *et al.* [8] searched underground at the Gran Sasso laboratory for the spontaneous electromagnetic radiation that collapse-driven charge diffusion would produce, found no excess, and tightened the bound on the effective nuclear mass-density size by about three orders of magnitude, ruling out the simplest DP model.

A non-Markovian gravitational channel is naturally *quasi-static*: with τ_c of order seconds the spectral weight of the noise sits at $1/\tau_c \sim 1$ Hz, while the spontaneous-radiation searches probe X-ray frequencies $\omega \sim 10^{19}$ s $^{-1}$. On energy-conservation grounds a direct-current-like noise carries no quanta capable of producing a keV photon, so one expects the radiation to be strongly suppressed.

We caution, however, that whether coloured (non-Markovian) collapse noise evades the spontaneous-radiation bound is an *open and contested* question, and we do not claim a clean evasion. Adler, Bassi and Donadi [9] find that the emission rate in collapse models is controlled by the *zero-frequency* component of the noise spectrum rather than by its value at the emitted-photon frequency, so that a high-frequency cutoff does not obviously reduce it; this is countered by calculations using normalizable final states and spatially bounded noise, in which the extra emission term vanishes [9, 10]. The ISP-Gaussian channel inherits exactly this unresolved issue, in common with other coloured-noise collapse models [10]. The defensible statement is therefore that the indivisible version is *plausibly* not excluded by Ref. [8] because its noise is quasi-static, but that settling this requires resolving the same zero-frequency-versus-emitted-frequency question that is open for coloured CSL.

Two consequences follow. First, the interferometric onset-shape test of Sec. V, not the radiation channel, is the decisive measurement, because radiation neither cleanly excludes

nor confirms the indivisible version. Second, the non-observation of collapse radiation mildly favours the large- τ_c (quasi-static, Gaussian) branch over the already-excluded Markovian edge.

IX. DISCUSSION: SCOPE AND LIMITATIONS

We state the boundaries of the result plainly. (i) *The energy scale is DP's, not new*: the contribution is the non-exponential shape and its falsifiable onset, not the magnitude $\tau_G = \hbar/E_G$, which is taken over from Refs. [1–3]. (ii) *The coefficient inherits DP's ansatz dependence*: whether the Gaussian time equals $\hbar/E_G$ exactly depends on the fluctuation amplitude σ_E ; only the quadratic shape is assumption-light. (iii) *The quadratic onset is generic to non-Markovianity*: it is shared by coloured-noise CSL and by non-Markovian environmental decoherence [10, 21, 22]; observing it would falsify *strict* Markovian DP/CSL and confirm a non-Markovian fundamental channel consistent with ISP, but would not by itself single out ISP among non-Markovian models. (iv) τ_c *is a genuinely new parameter*, fixed here only dimensionally (Sec. VII). (v) *Environmental non-Markovianity must be controlled*: because a non-Markovian environment can also produce a quadratic onset, the test requires that the residual after environmental subtraction be attributable to the gravitational channel—most cleanly demonstrated by its scaling with E_G at fixed environment.

Within these bounds, the result is a concrete, falsifiable, and motivated alternative to the Markovian DP law, derived from a specific and independently-motivated stochastic ontology [11, 12], and testable with the same platforms already pursuing DP [16, 17, 19, 24].

X. CONCLUSION

Treating gravitational decoherence as a process in an indivisible (non-Markovian) stochastic dynamics replaces the Diósi–Penrose exponential with a non-exponential, Gaussian-onset coherence decay at the same gravitational energy scale, with the Markovian DP law recovered in the zero-memory limit. The distinction is experimentally sharp and free of fitted constants: a linear onset of the residual gravitational log-coherence signals Markovian collapse, a quadratic onset signals the indivisible channel. The new time scale τ_c controlling the crossover is fixed dimensionally to the order of τ_G , placing the effect within reach

of mesoscopic-superposition experiments. The spontaneous-radiation bound that excludes parameter-free DP is plausibly evaded by the quasi-static non-Markovian noise, though this rests on an unresolved question common to coloured-noise collapse models. The onset-shape measurement is the decisive test, and it is accessible now.

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